

Ultrasonic Sensor Characterization
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INTRODUCTION

Sensor characterization is the process of testing and understanding how a sensor works by measuring important factors like accuracy, sensitivity, and response time. This helps ensure that the sensor gives reliable and correct data for its intended use. The process involves checking how the sensor reacts to different conditions and possible disturbances. By studying a sensor's behavior, we can improve its performance, make it more reliable, and ensure it works well in real-world situations.

Ultrasonic sensor characterization is the process of testing and understanding how an ultrasonic sensor works by measuring important factors like accuracy, sensitivity, and response time. This helps ensure the sensor provides reliable distance measurements for applications like obstacle detection and level sensing.

AIMS AND OBJECTIVES

- To measure the accuracy and precision of distance readings from the ultrasonic sensor.
- To analyze how different materials and surfaces affect the sensor's performance.
- To determine the sensor's response time and range limitations.
- To evaluate sensor performance at different reflection angles.

EXPERIMENTAL SETUP

1. Effective Range Test

- The ultrasonic sensor was tested by placing objects at various known distances.
- The sensor readings were recorded and compared to actual distances.
- The maximum effective detection range was identified.

2. Material Effect Test

- Three different materials (white wall, glass, and mirror) were placed at a fixed distance.
- The sensor readings for each material were recorded to analyze variations in reflection.

3. Reflection Angle Test

- Objects were positioned at different angles relative to the sensor.
- The impact of angular reflection on distance measurements was observed.

RESULTS

Effective Range Test

Distance (M)	Reading 1 (cm)	Reading 2(cm)	Reading 3(cm)	Average(cm)
1	101	101	101	101
2	202	202	202	202
3	304	305	306	305
4	406	407	406	406
5	0	0	0	0

Material Effect Test (Setup = 1m Distance)

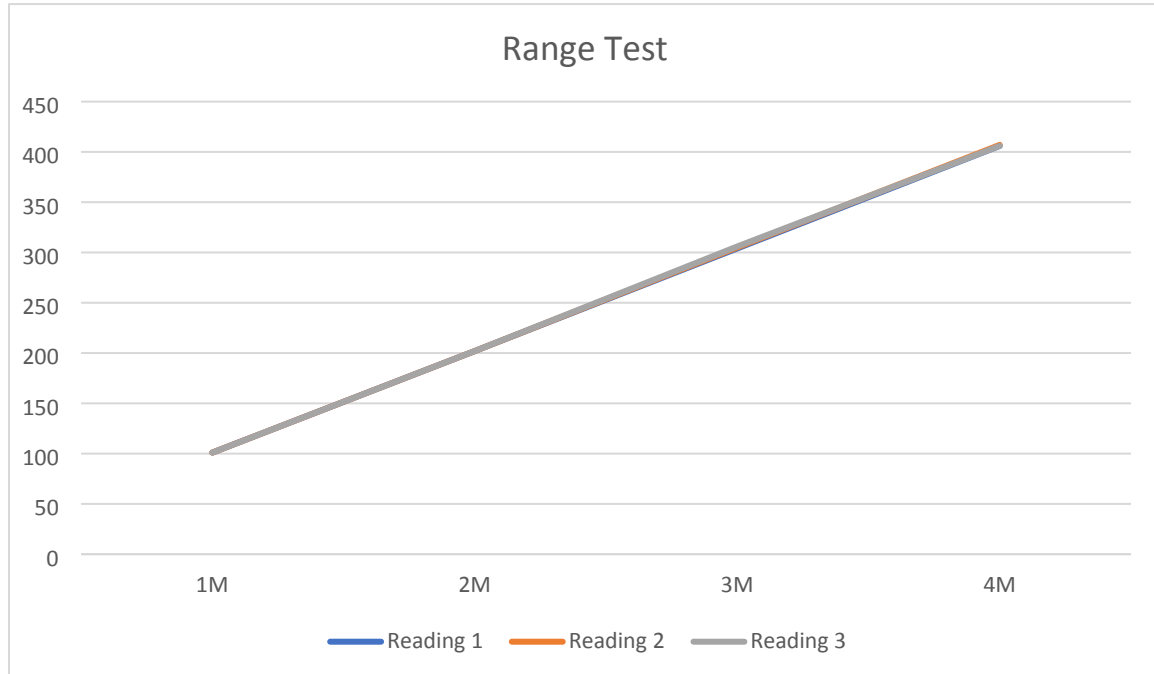
Material	Reading 1 (cm)	Reading 2 (cm)	Reading 3 (cm)	Average (cm)
White Wall	101	101	101	101
Glass	101	101	101	101
Mirror	102	102	102	102

Reflection Angle Test (Setup = 20cm Distance, material: mirror)

Angle	Reading 1 (CM)	Reading 2 (CM)	Reading 3 (CM)	Average (CM)
90	20	20	20	20
45	24	18	21	21
15	0	0	0	0

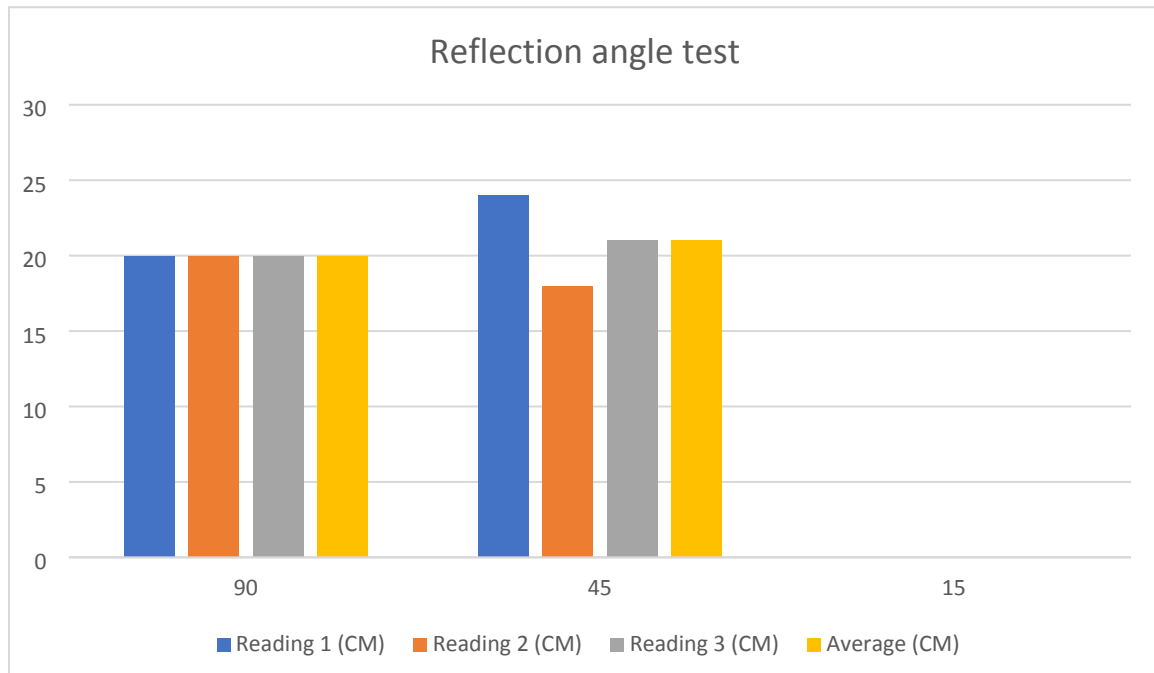
RESULTS ANALYSIS AND DISCUSSION

Effective Range Test: The sensor accurately detected objects up to 4 meters, but beyond this distance, it failed to provide any readings. This shows that the sensor has a limited range, making it suitable for short to medium-distance applications.



Material Effect Test: The sensor gave almost the same readings for white walls, glass, and mirrors. This means that different surfaces do not significantly affect the sensor's accuracy, as long as the surface is reflective to ultrasonic waves.

Reflection Angle Test: The sensor performed well when objects were directly in front (90°) but became less reliable at 45° and completely failed at 15°. This shows that ultrasonic sensors work best when the waves reflect directly back to the sensor. At steep angles, the waves are deflected away, leading to incorrect or no readings.



CONCLUSION

The ultrasonic sensor characterization tests showed that the sensor provides accurate distance measurements up to 4 meters but fails beyond this range. Different materials, such as white walls, glass, and mirrors, had minimal effect on the sensor's accuracy, confirming that it works reliably on reflective surfaces. However, the reflection angle test revealed that the sensor performs best when the waves return directly to it (90°) but becomes unreliable at steep angles. These findings help understand the sensor's limitations and can guide its use in real-world applications like obstacle detection and distance measurement.